

- LIVE AUTONOMOUS DIGITAL TWIN

Real moving autonomous ambulances with dynamic Time-Dependent A* routing, civilian yielding behavior, FIPA-ACL auctions, and bilateral hospital bed allocation.

Click
Node

Click Road

Click
Vehicle

● Autonomous Road Network Topology (Metropolitan Zone A)

CLEAR NIGHT



 **Spawn ESI-1 Cardiac Arrest**

⚠ Inject Traffic Jam (D* Lite)

 Toggle Weather

|| Pause

 Reset Fleet

Mean Response Time

5.4 min

Ambulance Offload Delay

1.8 min

911 Calls Handled

26

Critical Resuscitations

22

100% Specialty Match Rate

○ Rescue Alpha (Member 1 Unit)

Status: IDLE_PATROL • Battery: 96% • Speed: 56 km/h

AUTONOMOUS AGENT

SENSORS:
GPS (±0.8m) • TomTom Live Traffic API • OBD-II Telemetry • Patient ECG/SpO2

ACTUATORS:
Steering/Throttle • Emergency Strobes • FIPA-ACL CNP Transceiver

UTILITY GOAL:
 $U_i = -(\gamma \cdot \text{ETA} + \lambda \cdot \text{cov}) = -(5.4 \text{ min})$



> FIPA-ACL Live Agent Message Bus

[14:54:39] [CNP CONTRACT AWARD] Dispatcher awarded contract to Medic Bravo (ETA: 2.6 min).

[14:54:39] [HOSPITAL AUCTION] Initiating bilateral matching for patient with Cardiac Arrest / Ventricular Fibrillation.

[14:54:39] [CNP CONTRACT AWARD] Dispatcher awarded contract to Rapid Charlie (ETA: 1.5 min).

[14:54:39] [HOSPITAL AUCTION] Initiating bilateral matching for patient with Acute Ischemic Stroke (LV0).

[14:54:40] [STABLE MATCH LOCKED] Apex Heart & Vascular Hub pre-reserved resuscitation bay. Surcharge factor: 1.00x.

[14:54:40] [STABLE MATCH LOCKED] Valley Care Pavilion pre-reserved resuscitation bay. Surcharge factor: 0.90x.

RUBRIC
PRESENTATION
DECK

01 MAS
Architecture

02 PEAS
Formulation
(3M)

03 Environment
& Agent (3M)

04 TDA* &
Gale-Shapley
(3M)

05 Review
2 Plan
& Viva
(1M)

< Prev

STAGE
01 /
05

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Foundational Architecture • Multi-Agent Paradigm

Why Traditional Centralized CAD Fails vs Distributed Multi-Agent Systems

Solving combinatorial dispatch complexity and eliminating ambulance offload delay via decentralized edge intelligence.



Centralized CAD Bottlenecks

Single Point of Failure & Combinatorial Explosion

01 Combinatorial State Explosion $O(n!)$

Traditional Computer-Aided Dispatch executes monolithic linear integer programming. When simultaneous 911 calls surge during urban crises, computing globally optimal assignments takes minutes, causing fatal dispatch delays.

02 Ambulance Offload Delay (AOD / "Ramping")

Up to 40% of an urban ambulance fleet is stranded idling in hospital parking bays because destination ERs are saturated. Centralized CAD selects hospitals based on proximity alone, blind to real-time ER surgical capacity.

03 Extreme Information Asymmetry

Central dispatchers lack telemetry on sudden hospital cath-lab outages, while hospital trauma teams receive zero real-time biometric telemetry until the ambulance physically arrives at the triage door.



AURA-EMS Multi-Agent Breakthrough

Edge Intelligence, Bilateral Matching & Continuous Resilience

01 Parallel Contract Net Auctions (O(k) Scalability)

Dispatch tasks are partitioned into localized contract bids. Ambulance agents compute their own Time-Dependent A* trajectories in parallel, achieving real-time dispatch decisions in under 800 milliseconds.

02 Bilateral Two-Sided Matching (Gale-Shapley)

Hospitals are active autonomous agents that dynamically price intake capacity. Resuscitation bays and Cath labs are locked before the vehicle wheels roll, reducing offload delay from 45 min to under 2.4 min.

03 Dynamic D* Lite Incremental Replanning

When traffic congestion or accidents block urban corridors, moving ambulance agents incrementally repair their shortest-time path without re-evaluating the entire city graph, saving 85% CPU overhead.

Interactive Verification: Watch the live simulation canvas above execute this multi-agent bidding and bilateral bed-locking flow.

Trigger Live CNP
Auction Now

